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RESEARCH ARTICLE

Hands-Free UAV Control: Real-Time Eye Movement Detection Using EOG and LSTM Networks

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ABSTRACT Industry 4.0 has created a growing need for effective human–robot collaboration (HRC). As robots and humans work more closely together, efficient communication becomes essential for coordinating their actions seamlessly. While speech may seem like the obvious choice for communication, noisy factory environments can render it impractical. Additionally, workers often have their hands occupied with assembly tasks, making hand-controlled interfaces less practical for controlling robots. To address these challenges, this paper presents a novel, hands-free method for robot control using electrooculography (EOG) signals—specifically, eye movements and blinks—with unmanned aerial vehicles (UAVs) used as the demonstration platform. We developed a real-time system that captures EOG signals through a graphical user interface (GUI), allowing users to direct their gaze toward on-screen commands and confirm selections with blinks. A Long Short-Term Memory (LSTM) model was trained to detect gaze coordinates and blink events from the EOG data. For performance benchmarking, we implemented a vision-based video-oculography (VOG) system as a camera-based baseline and compared the two approaches under various conditions, including low light, wind disturbances, and deployment on embedded hardware as a resource-constrained device. Our comparative analysis showed the EOG-based method achieved 97.6% accuracy in command detection under normal conditions and maintained strong performance across challenging environments where the vision-based VOG system experienced significant degradation. The proposed EOG method also operated in real time at 47.97 FPS on a high-performance workstation and 38.87 FPS on the resource-constrained NVIDIA Jetson Orin Nano. On this platform, it ran 13.9× faster and used 319× less physical memory than the VOG-based alternative. These results underscore the advantages of EOG-based control in delivering reliable, real-time performance on both high-performance workstations and resource-constrained platforms, outperforming vision-based alternatives in efficiency and adaptability.

INDEX TERMS Electrooculography (EOG), video-oculography (VOG), human–machine interface (HMI), machine learning, long short-term memory (LSTM).

I. INTRODUCTION

Industry 4.0 has accelerated the shift from mass production to mass customization, driven by advancements in automation, robotics, and smart systems. This transition is transforming traditional manufacturing processes, placing a strong emphasis on human-robot collaboration (HRC) to meet the growing demands for customized products and flexible production lines. As robots and humans work together more closely, clear and efficient communication becomes crucial for ensuring smooth and coordinated interactions [1].

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While speech may seem like an intuitive form of communication, noisy factory environments can significantly hinder its effectiveness, often leading to errors and reduced efficiency. Even in cases where speech is available, it is not always the most efficient means of communication. For example, a human/robot that verbally announces every action, such as a handover, could slow down the workflow and interrupt the natural flow of interactions, especially in environments that require frequent exchanges between humans and robots. In such environments, relying solely on speech-based interfaces becomes impractical for precise and reliable robot control. Additionally, factory workers frequently engage in hands-on tasks, such as assembly or material handling, which limits their ability to use hand-controlled interfaces for directing robots. This constraint highlights the need for alternative control methods that do not rely on manual input, enabling workers to interact with robotic systems without interrupting their primary tasks.

One promising alternative to speech and hand-based interfaces is gaze control, which leverages eye movements to communicate commands to robotic systems. Eye movements are the coordinated actions that the eyes perform to scan the surroundings, focus on specific objects, or follow moving targets within the visual field. These movements are crucial for the accurate processing of the visual data, ensuring that images remain clear and stable, and for shifting attention to different parts of the visual scene. These movements can be classified into several categories, such as saccades (quick, jerky movements between points of focus), smooth pursuit (tracking a moving object), and fixations (when the eyes remain still, focused on a single point) [2].

Eye movement can be detected through several methods, each offering different levels of precision and suitability for various environments. Video-oculography (VOG) uses cameras to track eye movements by analyzing the position of the pupil, providing high accuracy in well-lit environments. Scleral search coil technology, which involves placing a small coil in the eye or on a contact lens, can measure eye movements with exceptional precision. However, this method requires the user to be positioned within a magnetic field, typically inside a specialized box or frame, making it more invasive and less practical for everyday use, especially in industrial settings. Electrooculography (EOG), on the other hand, detects eye movements by measuring the electrical potential difference between the cornea and retina. EOG is a non-invasive method, and since it measures biopotential signals, it remains unaffected by varying lighting conditions, making it highly reliable in industrial settings where light levels may fluctuate. This robustness, combined with its non-intrusive nature, makes EOG a strong choice for control applications, particularly in environments that require consistent performance across different conditions [3], [4].

Eye tracking technologies are widely used in fields such as human-computer interaction (HCI), assistive robotics, machine/robot control, cognitive load assessment, and stress detection. In [5], a multimodal approach was used to assess stress in a manufacturing environment by combining heart rate, galvanic skin response (GSR), and EOG signals collected using JINS EOG eye glasses [6]. Blink times were measured to extract features, which were then used to classify stress and no-stress conditions using a Support Vector Machine (SVM) classifier. Similarly, EOG, Electrodermal Activity (EDA), and Electroencephalography (EEG) were used in [7] to measure cognitive load in an industrial setting. In this case, blink rate was analyzed to extract features that indicated cognitive load, drawing parallels to the previous study, which focused on stress detection.

In [8], a multimodal approach integrating EOG, Electromyography (EMG), and EEG were used to control a soft robotic hand for intention recognition in rehabilitation applications. The system employed a thresholding algorithm to detect right, left, and double blink events, facilitating precise control of the exoskeleton. A similar approach was implemented in [9] where EOG, EMG, and EEG were used to control a robotic arm. In this case, the thresholding algorithm detected vertical, horizontal, and blink events to achieve control of the arm. A different multimodal approach was implemented in [10], using both EOG and EEG to control a soft robotic hand. Unlike the previous approach, this system did not incorporate blink detection but instead relied on recognizing left and right eye movements to drive the control mechanism. In [11], a single-channel EOG system was employed to detect horizontal eye movements using a thresholding algorithm, enabling control of a drone. Similarly, EOG signals were used to control a drone via a thresholding algorithm in [12]. The system employed imperceptible graphene-based electronic tattoos (GET) to detect eye movements—up, down, left, right, and blinks—which were translated into drone commands based on predefined voltage thresholds. In [13], a calibration phase was used to adjust the thresholding classifier by having users perform specific eye movements, such as up, down, left, and right. This calibration allowed the system to fine-tune the thresholding algorithm, ensuring accurate detection and classification of EOG signals. In [14], a single channel EOG system was developed for wheelchair control, specifically focusing on blink detection. The system presented users with a graphical user interface (GUI) that displayed several flashing buttons, each representing a different command, such as turning left or right. The buttons were highlighted one by one in a predefined sequence, and users issued commands by synchronizing their blinks with the button flashes. A thresholding algorithm was used to detect blinks from the vertical EOG signals, allowing the system to distinguish between target and non-target buttons. In [15], K-Nearest Neighbors (KNN), SVM, and Artificial Neural Networks (ANN) were employed to classify eye movements such as up/down and right/left for human-computer interaction. Similarly, [16] used Multi-Layer Perceptrons (MLP) to control an omnidirectional robot via eye movements, while

outperforming thresholding methods, these classifiers do not account for the temporal aspects of EOG signals.

Considering that eye movements, especially blinks, involve varying intensities and are primarily characterized by their temporal dynamics, thresholding methods and traditional classifiers such as KNN, SVM, and MLP often fall short in efficiently capturing these nuances. Hence, LSTM networks, with their ability to model temporal dependencies, provide a more effective solution for handling these subtle variations. Building on this, we present a hands-free, real-time approach for UAV control that leverages LSTM networks to enhance the accuracy and reliability of eye movement and blink detection using EOG signals. Our framework involves the collection of an EOG dataset, preprocessing the data using wavelet transforms, and applying a dual-output LSTM model for both coordinate tracking regression and blink classification tasks. To highlight the advantages of our proposed LSTM-based EOG system over the existing VOG alternative, we implemented a vision-based VOG system by detecting facial landmarks using Dlib-ML [17] and localizing the iris using YOLOv8 [18]. We evaluated the performance of our LSTM-based EOG method against VOG under various conditions—normal, low light, and windy—using both a high-performance workstation and a resource-constrained NVIDIA Jetson Orin Nano [19]. This allowed us to compare both techniques effectively across different environmental and hardware contexts.

The key contributions of this paper are:

- Design of a data collection scenario with automatic blink labeling, capturing both saccadic eye movements and blinks for the LSTM model training,
- Development of a dual-output LSTM architecture that effectively processes EOG signals for simultaneous gaze coordinate prediction and blink detection, leveraging the temporal dynamics of eye movements,
- Implementation of a multi-threaded parallel computing system to process EOG signals through the LSTM model efficiently, enabling real-time analysis and response to eye movements,
- Comparative evaluation of our LSTM-based EOG gaze tracking method against VOG techniques across various environmental conditions (normal, low light, and windy), highlighting the robustness of EOG signals in challenging environments, and
- Performance assessment of the LSTM-based system on both high-performance workstations and resource-constrained NVIDIA Jetson Orin Nano platforms, with practical demonstration of UAV control using the embedded implementation.

The rest of the paper is organized as follows: Section II describes the methodology, detailing the data collection process, signal preprocessing, the development of the LSTM-based EOG architecture with its multi-threaded parallel computing implementation, and the VOG model architecture. The experimental results and system performance are discussed in Section III, including comparisons of the LSTM-based EOG method with the VOG under various environmental conditions and on different computational platforms. Section IV provides a comprehensive discussion of the findings, addressing the implications of our results, contextualizing them within the existing literature, identifying both strengths and limitations of the proposed approach, and suggesting avenues for future work. Finally, Section V concludes the paper.

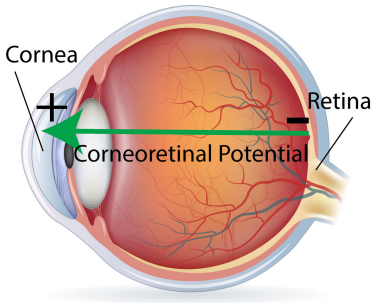


FIGURE 1. Corneoretinal potential (©reineg/Adobe Stock).

II. METHODS

In this study, our primary objective was to design an interface capable of mapping a user's gaze to specific x and y pixel coordinates on a screen. This mapping enables the detection of gaze fixation on particular commands, which can then be sent to the DJI Tello drone [20] for execution. To achieve this, we decided to leverage machine learning techniques to accurately detect both gaze coordinates and blink events. Recognizing the importance of accurate data for machine learning, we first collected a comprehensive dataset of EOG signals. This dataset was then used to train an LSTM model, chosen for its ability to handle temporal dependencies in sequential data. The trained model was subsequently employed to detect gaze coordinates and blink events in real time, which allowed for the precise execution of drone commands. Each of these steps—data collection, model training, and system implementation—will be explained in detail in the following sections.

A. LSTM-BASED EOG METHOD

1) EOG DATA ACQUISITION

EOG, a technology that originated in the 1920s, takes advantage of the electrical potential difference between the cornea, which has a positive charge, and the retina, which carries a negative charge. This difference, known as the corneoretinal potential (as depicted in Fig. 1), allows the eye to act as an electric dipole [3], [4]. This creates an electric field $\mathbf{E}$ around the eye, as described by Coulomb's law [21]:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r} \tag{1}$$

where q is the charge, r is the distance from the charge, and $\hat{r}$ is the unit vector along the radial direction. As the eye moves, the orientation of this dipole changes, resulting in a time-varying electric field $\mathbf{E}(t)$. According to the

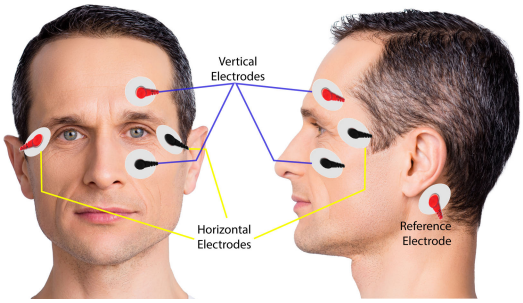


FIGURE 2. EOG vertical and horizontal configurations (©deagreez/Adobe Stock).

Ampère-Maxwell Law [22] given in (2), a changing electric field induces a magnetic field **B**:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{2}$$

Here, **B** is the magnetic field, **J** is the current density, and $\frac{\partial \mathbf{E}}{\partial t}$ represents the rate of change of the electric field over time. The magnitude of the induced magnetic field depends on how quickly the electric field changes, which is directly related to the speed of eye movement.

Furthermore, Faraday's Law of Induction (3) states that a changing magnetic field **B** induces an electric field ($\mathbf{E}_{\text{induced}}$) [22]:

$$\nabla \times \mathbf{E}_{\text{induced}} = -\frac{\partial \mathbf{B}}{\partial t} \tag{3}$$

This induced electric field generates a voltage on conductive surfaces, such as the skin, which is detected by the EOG electrodes. The voltage measured by EOG electrodes on the skin surface near the eye is therefore influenced by the dynamics of the eye movement.

Electrodes are typically made from silver (Ag), silver chloride (AgCl) [13], [23], or gold [24], which are the most widely used commercially available electrodes for biopotential measurements, mainly due to their low skin-electrode impedance and minimal DC drift [23], [24]. These electrodes offer reliable signal acquisition and are cost-effective. For horizontal EOG measurements, electrodes are typically placed on the outer canthus of each eye, while for vertical EOG recordings, they are positioned above and below the eye. The electrodes are referenced to a neutral site, usually over the mastoid region behind the ear (Fig. 2) [25], [26], [27].

EOG signals typically have low amplitudes, ranging from 50 to 3500 μV [13], [25], which necessitates amplification for accurate processing. These signals span a frequency range from DC to 50 Hz and are prone to noise from various sources, including electrode contact noise, power line interference, intrinsic electronic noise, motion artifacts, and muscle contractions (EMG) [13], [25], [28]. To address these issues, amplification and filtering are essential to improve signal quality. Proper filtering and noise cancellation techniques allow the raw EOG signal to be separated from noise components. Typically, a notch filter [28] or low-pass filter (LPF) [13] is used to minimize 50/60 Hz power line noise, while a high-pass filter (HPF) [13], [24] or Driven Right Leg (DRL) [14], [24] circuit is employed to remove the skin-electrode DC offset. The DRL circuit helps reduce common-mode noise by feeding the common-mode voltage back into the body, which enhances the system's ability to reject interference and improves overall signal quality. Generally, both filtering and amplification are managed through biopotential data acquisition or Analog Front-End (AFE) systems at the hardware level. In this study, we utilized two BioAmp EXG Pill AFE biopotential signal acquisition boards [29], each equipped with an amplifier with a gain of 50 dB, a bandpass filter, and a DRL circuit to capture EOG signals. After amplification and filtering through the AFE, the analog EOG signals were digitized using a 10-bit ADC and transmitted to the computing platform via USB serial communication at 9600 bits/s for further processing (Fig. 3).

To map EOG signals to the corresponding x and y pixel coordinates and record blink data, we designed a GUI that begins with a five-second countdown, allowing the user to focus their gaze at the center of the screen. After the countdown, a circular pointer appears at the center and remains there for two seconds. The pointer then disappears and reappears at a random position on the screen, where it stays for another two seconds before returning to the center. This cycle repeats for a total duration of 60 seconds. To ensure that the user's eye movements closely match the pointer's positions, we included a visual guide in the form of a line connecting the previous pointer location to the next one. This prevents the user from searching for the pointer, which could cause the recorded EOG data to misalign with the actual pointer positions (Fig. 4).

Fig. 5 shows a sample of the collected vertical EOG data, including the normalized raw signal (blue), and normalized pointer y-coordinates (green). In the vertical channel, upward eye movements produced positive deflections while downward movements generated negative deflections. Blinks created distinctive high-amplitude biphasic patterns (positive-negative spikes) that were clearly distinguishable from regular eye movements.

Fig. 6 presents the corresponding horizontal EOG data with normalized raw signal (blue), and normalized pointer x-coordinates (green). The horizontal channel primarily captured leftward and rightward eye movements while showing minimal response to blinks or vertical eye movements. This characteristic difference between channels allowed us to isolate blink detection in the vertical channel while using both channels for gaze coordinate estimation.

As the users were instructed to blink when the pointer was stationary, we could accurately label these intentional blink events using our automatic blink labeling system. Fig. 7 illustrates this process in sequential steps. First, we identified state changes in the pointer movement data, with rising edges representing transitions from lower to higher amplitude (upward pointer movement) and falling edges representing transitions from higher to lower amplitude

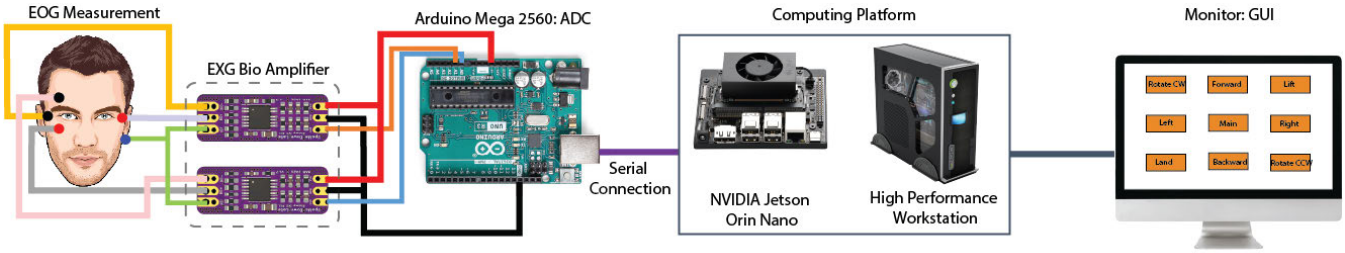


FIGURE 3. Hardware setup for real-time LSTM-based EOG gaze tracking.

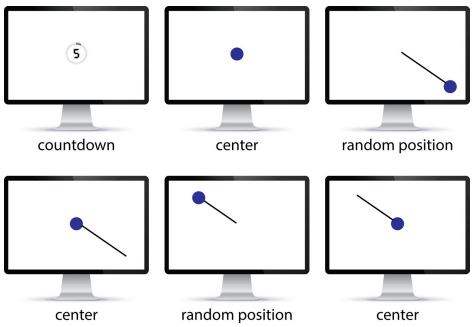


FIGURE 4. Dataset collection GUI (©Can Yesil/Adobe Stock, ©ellyson/Adobe Stock).

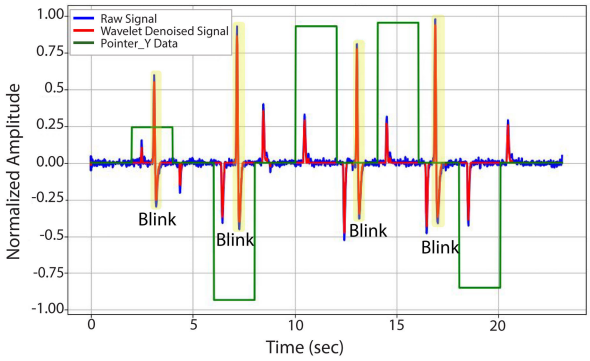


FIGURE 5. Vertical EOG with denoised signal and pointer y-coordinates.

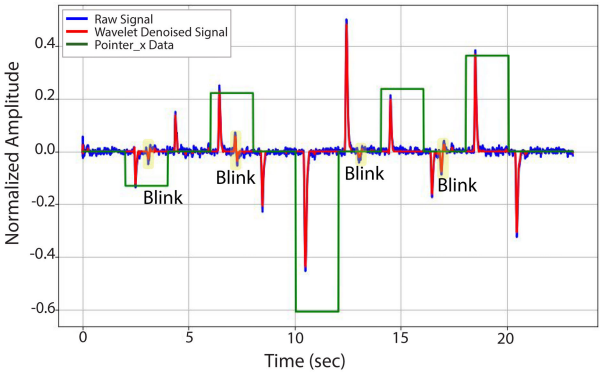


FIGURE 6. Horizontal EOG with denoised signal and pointer x-coordinates.

(downward pointer movement), as shown in Fig. 7-(a). We then matched these edges with corresponding peaks in the vertical EOG signal (Fig. 7-(b)). By identifying the closest positive peak following each rising edge and the closest negative peak following each falling edge within a calculated window (one quarter of the duration between consecutive edges), we isolated the eye movement components. For each identified peak, we located the zero crossings immediately before and after the peak and removed these segments from the original signal (Fig. 7-(c)), effectively extracting the entire eye movement component while preserving potential blink events. The resulting signal predominantly reflected blink activities, with the cleanly isolated peaks automatically labeled as '1' for blinks and '0' for non-blink intervals (Fig. 7-(d)).

After each collection session, EOG signals were visually examined for abnormal behavior or artifacts to validate data quality before proceeding with model training.

To ensure real-time data collection, we recorded the EOG signals, pointer positions, and time information using two parallel threads, maintaining synchronization between the eye movement data and the pointer location. The final input data for training the model consisted of horizontal and vertical EOG signals, and labels formatted as (x, y, blink). Here, x and y represent the pointer coordinates, while blink is a binary indicator. During blink events, x and y values were set to zero, indicating no saccadic movement, resulting in (0, 0, 1) for blink events. For saccadic movements without blinks, the coordinates were set to (x, y, 0). When neither a blink nor saccadic movement occurred, the label was (0, 0, 0). After completing the data collection process, we gathered a total of 439,811 samples of EOG data corresponding to 1,500 target points on the screen.

2) PREPROCESSING

Although the EOG data underwent initial filtering during the hardware stage, additional signal processing in the software stage was suggested by the literatures [13], [25], [30], and [31]. Our preprocessing pipeline first normalized the raw EOG signals to the range $[-1,1]$. Following normalization, we implemented the discrete wavelet transform (DWT) method [32] for further denoising as it effectively captures localized temporal variations in the signal. The DWT decomposes a discrete-time signal into approximation and detail coefficients across multiple levels, enabling multiresolution analysis in both time and frequency domains.

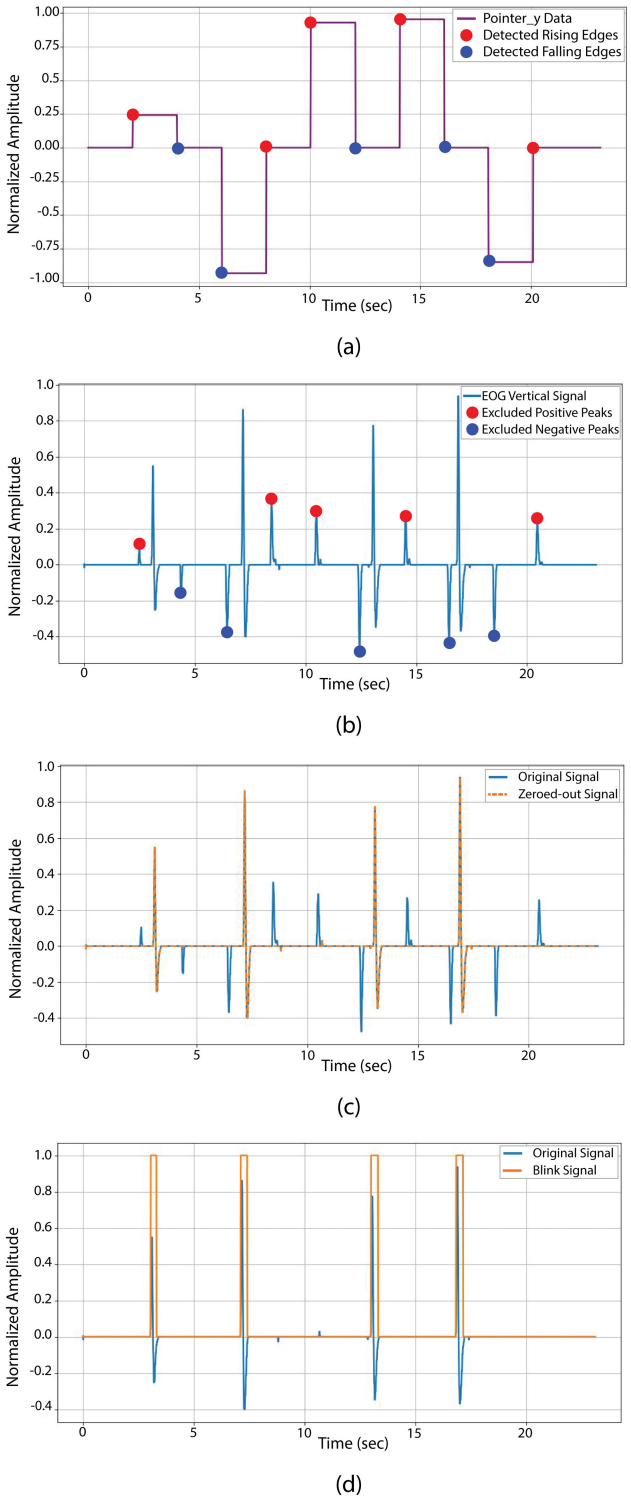


FIGURE 7. Sequential steps in the automatic labeling of blink events.

Mathematically, the one-level DWT of a signal $x[n]$ using low-pass filter $h[n]$ and high-pass filter $g[n]$ is given by:

$$A[k] = \sum_n x[n] \cdot h[2k - n], \quad D[k] = \sum_n x[n] \cdot g[2k - n] \tag{4}$$

where $A[k]$ and $D[k]$ represent the approximation and detail coefficients at level 1, respectively, with $k = 0, 1, \ldots, \lfloor N/2 \rfloor - 1$. The high-pass filter $g[n]$ is derived from the low-pass filter through the quadrature mirror relationship $g[n] = (-1)^n h[1 - n]$. This convolution followed by downsampling by a factor of 2 reduces the coefficient length to approximately half that of the original signal, enabling efficient multiresolution decomposition while preserving essential signal characteristics for effective denoising.

For our implementation, we selected the Symlet20 (Sym20) wavelet at decomposition level one due to its near symmetry and higher number of vanishing moments compared to simpler wavelets. The wavelet denoising was applied on a rolling window of the most recent 50 samples, enabling real-time signal cleaning while maintaining temporal integrity critical for accurate eye movement and blink detection.

Since the eye naturally followed the pointer with a slight delay, the EOG signals lagged behind the pointer positions. Recognizing this misalignment between EOG signals and pointer positions, we implemented a temporal shift to correct this discrepancy. Following this, we segmented the data into time-series segments using a sliding window approach, creating the necessary input sequences for our LSTM model.

Due to the two-second fixation periods and the much shorter durations of saccadic movements, the dataset became imbalanced, resulting in a larger volume of data representing fixation events. Additionally, the infrequent occurrence of blink events further contributed to this imbalance. To rectify this, we randomly removed segments containing fixation events, ensuring that the LSTM model could effectively learn from a balanced representation of fixation, saccadic, and blink events. Subsequently, we divided the data into training, validation, and test sets with a ratio of 70/15/15.

3) LSTM NETWORK TRAINING

Given that EOG signals are time-sequence data, we opted to use an LSTM network [33] for our machine learning model, as LSTMs are specifically designed to handle sequential information and capture temporal dependencies. These networks are structured around memory cells that are controlled by three gates: the input gate, the forget gate, and the output gate, which work together to manage the flow of information through the network, ensuring that essential information is retained while irrelevant data is discarded.

Building on this foundation, we designed our LSTM model with three-layers, where each layer contained 128 units (Fig. 8). A dropout rate of 0.3 was incorporated into the model to prevent overfitting. Our model utilized two output heads: one for regression and the other for classification. The regression head, implemented as a linear layer without an activation function, predicted the continuous x and y coordinates of the pointer. The classification head used a sigmoid activation function to produce a probability output, predicting whether a blink event has occurred. To train

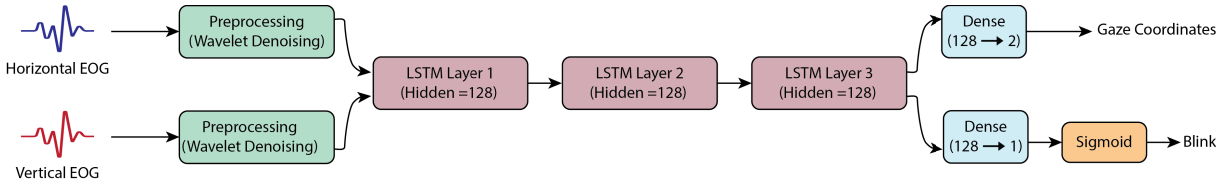


FIGURE 8. Model diagram of the proposed LSTM network with regression and classification heads.

this dual-output model using the Backpropagation Through Time (BPTT) method, two separate loss functions were utilized: Mean Squared Error (MSE) for the regression task (5), to minimize the difference between predicted and actual pointer coordinates, and Binary Cross-Entropy (BCE) for the classification task (6), to ensure accurate detection of blink events [34]. The MSE and BCE are defined as:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{5}$$

$$\text{BCE} = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{6}$$

where y_i is the actual gaze coordinate and $\hat{y}_i$ is the predicted gaze coordinate.

The total loss for optimization was a combination of these two, balancing the regression and classification objectives to ensure that both pointer tracking and blink detection were effectively learned. For optimization, we utilized the Adam optimizer [35] with a learning rate of 0.001, training the model for 50 epochs with a batch size of 16. Adam's adaptive moment estimation adjusts the learning rates of individual parameters based on the gradients' first and second moments, which facilitates efficient convergence during training.

The computational complexity of the proposed pipeline comprises two main components: preprocessing and inference. The preprocessing stage, which involves discrete wavelet-based denoising applied to a fixed-length window of EOG data across n channels, has a complexity of $\mathcal{O}(nT)$, where T denotes the window length and n is the input dimension or the number of channels.

For the LSTM-based inference, each layer incurs a per-timestep complexity of $\mathcal{O}(nh + h^2)$, where h is the number of hidden units and n is the input dimension. Given a sequence length of T, the total inference complexity across all timesteps is $\mathcal{O}(T(nh + h^2))$.

Therefore, considering both the wavelet-based preprocessing and the sequential inference through the LSTM layers, the total complexity per input sequence should be:

$$\mathcal{O}(T(nh + h^2)) \tag{7}$$

which supports the model's applicability to continuous, real-time signal interpretation with minimal delay.

4) REAL-TIME INFERENCE PIPELINE

After training the LSTM model offline for gaze coordinate regression and blink classification, we integrated it into a real-time system for EOG-based gaze detection. To ensure responsive and low-latency performance, our implementation adopted a parallel computing approach, where multiple threads operate concurrently to handle distinct stages of data processing. Specifically, one thread was dedicated to continuously receiving EOG data and queuing it for processing. Simultaneously, another thread retrieved this data from the queue and applied a wavelet filter for denoising. Meanwhile, a third thread ran the trained LSTM model to process the filtered data, assigning the x and y coordinates and blink labels. This parallel thread structure ensures efficient data handling, allowing for seamless real-time mapping of EOG signals to control commands. Once the x and y coordinates of the gaze are determined, the system identifies the closest command to the pixel within the GUI (Fig. 9), representing the user's intended gaze point.

B. VOG METHOD

In this study, we implemented a VOG system to conduct a comparative analysis with EOG techniques for gaze tracking. The VOG setup involved positioning a standard webcam at the bottom of the monitor, oriented towards the participant's face. This camera continuously captured video frames, enabling real-time facial and eye region detection (Fig. 10).

Using the Dlib-ml computer vision library [17], we accurately identified 68 facial landmarks, including periocular points. These points were then connected via a B-spline interpolation algorithm to create a precise contour of the ocular region. Subsequently, we extracted the eye areas, ensuring sufficient margins for complete capture of all relevant ocular features. These regions were then processed with a YOLOv8 [18] nano segmentation model, specifically fine-tuned on the I-SOCIAL-DB dataset [36] for accurate iris segmentation. Once the iris was detected, the system first assessed the blink status using an Eye Aspect Ratio (EAR)-based method [37]. If the eyes were determined to be open, the system proceeded to estimate horizontal and vertical gaze directions based on the relative position of the iris within the eye region (Fig. 11).

1) BLINK DETECTION

To detect blinks, we utilized the EAR [37], a geometric-based approach that quantifies eye openness using periocular

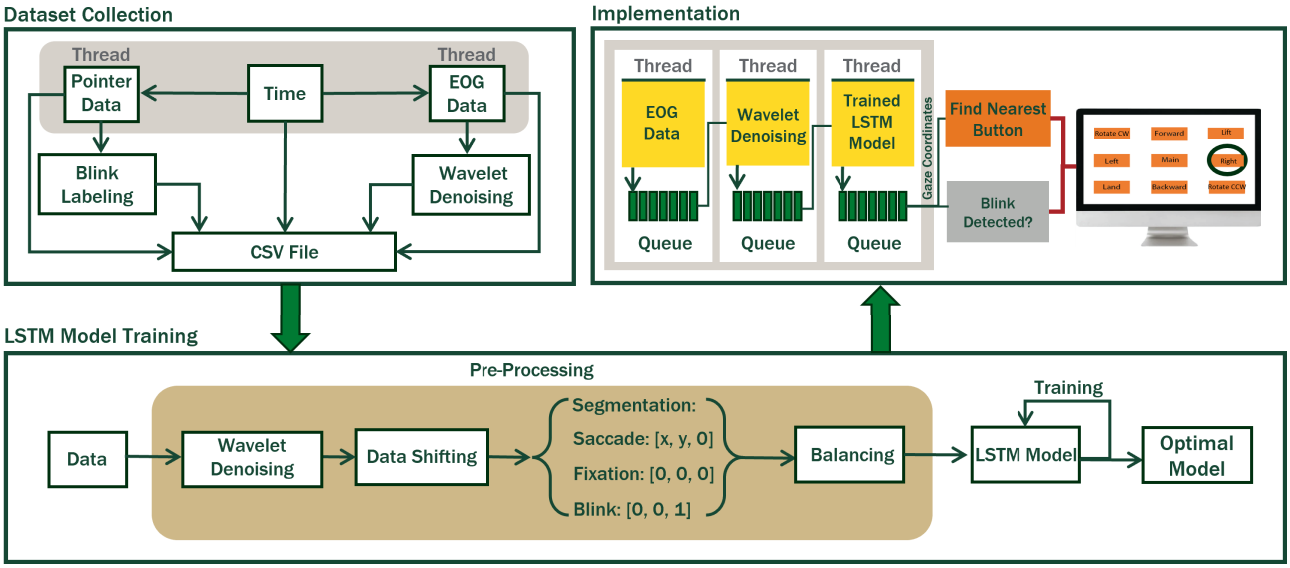


FIGURE 9. Overall framework for real-time LSTM-based EOG gaze tracking.

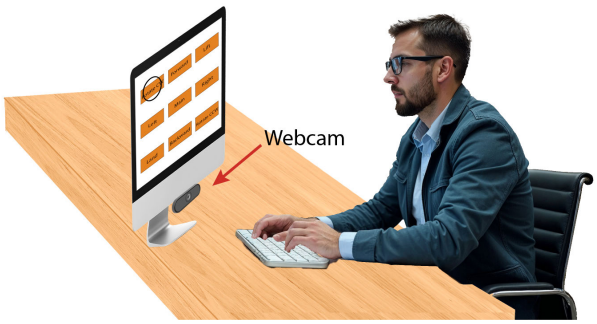


FIGURE 10. VOG hardware setup (©abu/Adobe Stock).

landmarks. The EAR is computed using six key eye landmarks detected by Dlib-ml, measuring the Euclidean distances between the vertical and horizontal eye points. It is defined as:

$$EAR = \frac{\|P_2 - P_6\| + \|P_3 - P_5\|}{2\,\|P_1 - P_4\|} \tag{8}$$

where P_1 to P_6 represent specific eye landmarks, with P_1 and P_4 defining the horizontal eye span, while P_2, P_3, P_5, P_6 correspond to the upper and lower eyelid positions (Fig. 12).

A blink event was identified when the EAR value dropped below a predefined threshold T_{blink} for a short duration, indicating full eyelid closure.

2) HORIZONTAL GAZE DETECTION

To determine horizontal gaze direction, we developed a ratio-based approach that leverages the relative positions of the iris centers and eye canthi. For each eye, we established a vertical line through the iris center that intersects with the horizontal line connecting the inner and outer canthi. This intersection point, termed the "iris projection point (IPP)", serves as a reference for distance calculations (Fig. 13).

For a given eye with iris center coordinates $(x_{\text{iris}}, y_{\text{iris}})$, inner canthus $(x_{\text{inner}}, y_{\text{inner}})$, and outer canthus $(x_{\text{outer}}, y_{\text{outer}})$, we computed the following distances:

$$d_{\text{inner}} = \sqrt{(x_{\text{IPP}} - x_{\text{inner}})^2 + (y_{\text{IPP}} - y_{\text{inner}})^2} \tag{9}$$

$$d_{\text{outer}} = \sqrt{(x_{\text{IPP}} - x_{\text{outer}})^2 + (y_{\text{IPP}} - y_{\text{outer}})^2} \tag{10}$$

Using these distances, a normalized ratio for each eye was defined as:

$$R_{\text{eye}} = \frac{d_{\text{inner}}}{d_{\text{outer}}} \tag{11}$$

This ratio characterizes the iris position relative to the eye width, accounting for individual morphological variations.

Although ratio-based approaches based on iris center geometry have been proposed [38], their sensitivity proved insufficient in our setup, leading to frequent misclassification between horizontal gaze states. To address this, we computed the gaze ratio separately for both eyes and introduced an interocular ratio, defined as the ratio of the right-eye value to the left-eye value.

$$R_{\text{interocular}} = \frac{R_{\text{right}}}{R_{\text{left}}} \tag{12}$$

This approach exploits the physiological nature of conjugate eye movements. When gazing rightward, the right eye's iris moves toward the outer canthus, decreasing R_{right}, while the left eye's iris moves toward the inner canthus, increasing R_{left}. As a result, the numerator becomes smaller while the denominator increases, yielding a distinctly small interocular ratio. Conversely, leftward gaze produces a large interocular ratio. This divisional method effectively amplifies subtle

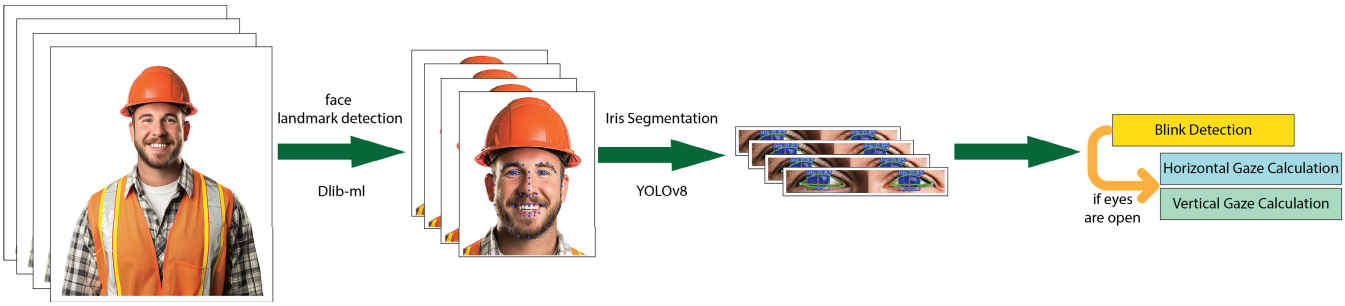


FIGURE 11. Overall framework for VOG gaze tracking (©Aitch/Adobe Stock).

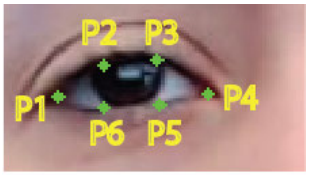


FIGURE 12. Visualization of eye landmarks.

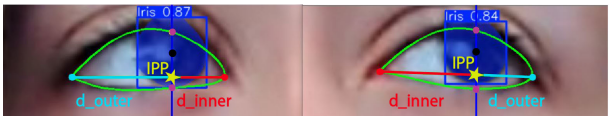


FIGURE 13. Horizontal gaze estimation.

eye movements, and normalizes for individual anatomical variations.

The final gaze classification was determined by comparing the interocular ratio against empirically established thresholds:

$$\text{Gaze Direction} = \begin{cases} \text{Right} & \text{if } R_{\text{interocular}} < T_{\text{right}} \\ \text{Left} & \text{if } R_{\text{interocular}} > T_{\text{left}} \\ \text{Center} & \text{otherwise} \end{cases} \quad (13)$$

where T_{right} and T_{left} are classification thresholds determined through calibration procedures.

3) VERTICAL GAZE DETECTION

To improve robustness in vertical gaze detection, we employed an area-based approach focused on visible scleral space, rather than relying on distances between the iris center and facial landmarks as in prior studies [38]. In this method, we first identified the right and left borders of the iris bounding box and extended them until they intersected with the previously computed B-spline contour of the eye. This process defined a closed region, where the upper boundary was formed by the line connecting the inner and outer canthi, the lateral boundaries were determined by the extended iris borders, and the lower boundary was shaped by the B-spline curve (Fig. 14).

Within this region, we computed the total iris area and specifically measured the portion of the iris located below the canthi line. The vertical gaze direction was then inferred by calculating the ratio of the iris area below the canthi line to the total bounded region. A ratio close to zero indicated an upward gaze, as the iris remained predominantly above the canthi line. A moderate ratio within a calibrated threshold corresponded to a forward gaze, while a high ratio, reflecting significant overlap of the iris with the lower bounded region, indicated a downward gaze.

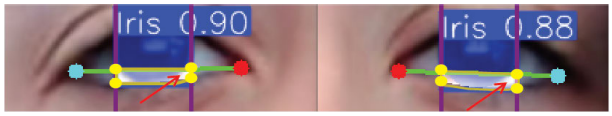


FIGURE 14. Vertical gaze estimation.

C. GAZE-BASED UAV COMMAND EXECUTION

Upon detecting the user's gaze direction—whether via EOG signals processed by the LSTM model or VOG-based iris tracking—the system mapped the gaze to a corresponding command button displayed on the user interface. To mitigate the risk of accidental activations caused by unintentional gaze fixation, a confirmation mechanism was employed.

When the user maintained their gaze on a specific command for three consecutive seconds, the system triggered a text-to-speech prompt requesting confirmation. This mechanism ensured that commands were only executed when the user was deliberately focusing on a control element, addressing potential ambiguities in gaze-based interaction.

If the user intended to proceed, they confirmed the command by performing a deliberate blink. Upon confirmation, the system verbally acknowledged the input, e.g., "Activating the command", and the DJI Tello drone [20] proceeded to execute the associated action (Fig. 15).

III. RESULTS

A. LSTM MODEL PERFORMANCE ON EOG DATA

In evaluating the performance of our LSTM model, we examined several key metrics that are critical to the success of our application: precision, recall, F1-score, and accuracy for blink classification [39], along with MSE, Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 [40] for coordinate tracking regression tasks (Table 1).

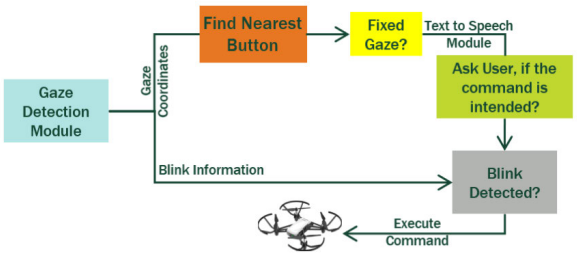


FIGURE 15. Gaze-based UAV control workflow.

TABLE 1. Performance metrics for blink classification and gaze coordinate tracking.

Classification Metrics				
Loss	Accuracy	Precision	Recall	F1-Score
0.04	0.99	0.98	0.97	0.98
Regression Metrics				
MSE	RMSE	MAE	R-Squared	
0.01	0.11	0.05	0.91	

In our system, precision emerged as the most relevant metric for ensuring that our UAV system responds only to intentional blinks. Defined as the proportion of true blink events among all predicted blink events (14), precision is crucial for preventing unintended commands, which could result from false positives.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{14}$$

where True Positives (TP) refer to instances when blink events are correctly detected, and False Positives (FP) occur when non-blink events are mistakenly identified as blinks.

In addition to precision, we considered recall, which captures the proportion of actual blink events that were correctly identified by the model. Recall is defined as:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{15}$$

where False Negatives (FN) represent cases when actual blink events are missed and not detected by the system.

Although recall is important for ensuring that most blinks are detected, it is less critical in our context since a missed blink can be repeated. However, high recall (0.97) complements our precision, ensuring that the system remains responsive.

The F1-score, given in (16), balances precision and recall, providing a single metric to assess the model's overall performance.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{16}$$

In our case, a high F1-score of 0.98 confirms that our model effectively handles both detecting blinks and avoiding false positives.

Finally, accuracy offers an overall view of the model's performance, defined as:

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}} \tag{17}$$

While accuracy is high (0.99), it may not fully reflect performance in cases of class imbalance, which is why precision and recall provide more insight into the model's effectiveness in our specific scenario.

For the regression component of our model, we assessed how well the model predicts the continuous x and y coordinates of the gaze using MSE, RMSE, MAE, and R^2 performance metrics. MSE, given in (5), measures the average squared difference between predicted and actual values, penalizing larger errors more significantly. RMSE, as the square root of MSE, provides an error measure in the same units as the original data, offering an intuitive understanding of error magnitude. Both metrics are valuable for minimizing large deviations from the true values.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{18}$$

In our results, the model achieved an MSE of 0.01 and an RMSE of 0.11, reflecting the model's low error rates and effective handling of larger deviations. The MAE metric, shown in (19), focuses on the average magnitude of errors without heavily penalizing larger deviations. With an MAE of 0.05, the model's predictions deviate minimally from the actual values, demonstrating strong predictive accuracy.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{19}$$

Finally, R^2 represents the proportion of variance in the gaze coordinates that is captured by the model, offering insight into the model's effectiveness in accounting for overall variability in the data. R^2 is calculated using:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{20}$$

The R^2 value of 0.91 demonstrates that the model has 91% of the variance in the data, indicating a strong fit to the observed data. This high R^2 score further supports the model's effectiveness in capturing the underlying patterns within the dataset.

B. COMPARATIVE RESULTS OF LSTM-BASED EOG VERSUS VOG FOR COMMAND AND BLINK DETECTION

To assess the effectiveness of biopotential and vision-based gaze tracking, we compared our LSTM-based EOG system with a VOG implementation in classifying gaze-based commands and detecting blinks under varying operational conditions. While the previous subsection focused on the regression accuracy of the LSTM model for continuous gaze estimation, here we evaluate its classification performance in

the context of command selection and compare it directly with the VOG-based method. This analysis highlights the practical strengths and limitations of each approach, offering insights into their applicability in real-world human-robot interaction (HRI) scenarios.

We developed a systematic testing protocol to evaluate both approaches. The testing procedure consisted of command selection and blink detection tasks. For command selection, a large circle would appear randomly over a command on the GUI, and successful detection occurred when the eye gaze tracked by either EOG or VOG appeared at the corresponding location. For blink detection, the system prompted users to blink, and successful detection was recorded when the system correctly identified the blink event. Each test comprised 50 iterations, and we conducted five separate trials for each testing scenario.

Both EOG and VOG systems were evaluated under three environmental conditions: normal operating conditions, windy conditions, and low-light conditions. Tests were conducted on both a high-performance workstation and a resource-constrained edge computing platform. The high-performance workstation was equipped with an Intel Xeon Gold 6238R CPU (2.20GHz), 192GB RAM, and an NVIDIA RTX A5000 GPU (24GB VRAM) running CUDA 12.3. For edge computing evaluation, we used an NVIDIA Jetson Orin Nano developer kit [19] featuring NVIDIA Ampere architecture with 1024 CUDA cores and 32 tensor cores delivering up to 67 INT8 TOPS of AI performance, a 6-core Arm Cortex-A78AE v8.2 64-bit CPU, 8GB 128-bit LPDDR5 memory with 102 GB/s bandwidth and power consumption ranging from 7W to 25W. The NVIDIA Jetson Orin Nano system ran NVIDIA JetPack 6.2 with CUDA 12.6 support.

For the environmental variations, low-light conditions were tested at illumination levels of 0.4, 1.6, and 27 lux. Windy conditions were simulated using a Honeywell Turboforce Power+ fan operating at maximum speed. Under normal conditions, users were seated approximately 24-25 inches away from the monitor, with eye distance from desk at 16 inches vertically. The camera was placed below the desktop at 9 inches from the desk vertically, with ambient room lighting at 70 lux. The monitor had physical dimensions of 20.7 × 11.7 inches, with the GUI containing nine commands divided into three columns and three rows equally distributed both horizontally and vertically.

1) NORMAL OPERATING CONDITIONS

Under normal operating conditions (70 lux ambient lighting, no wind disturbance), we evaluated the performance of both EOG and VOG systems for command selection and blink detection on a high-performance workstation. This baseline scenario allows for a direct comparison of the inherent capabilities of each approach without environmental confounding factors. The results demonstrate the relative strengths of each system when operating in ideal conditions, providing context for understanding performance degradation in more challenging environments.

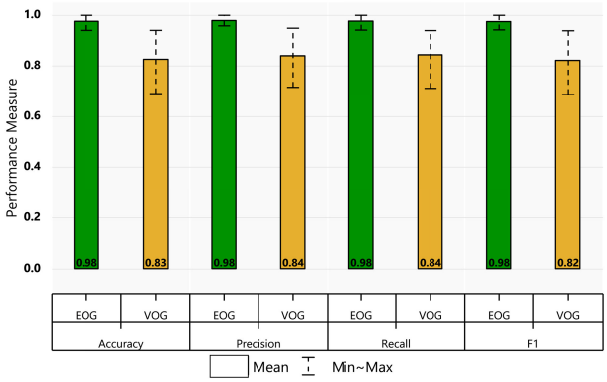


FIGURE 16. Command selection comparison under normal condition.

For command selection, the LSTM-based EOG system achieved consistently high performance across all metrics (Fig. 16). Over five test iterations, the EOG system maintained an average accuracy of 97.6% (SD=1.9%), with precision and recall values of 98.2% and 97.9% respectively. The F1-score, which balances precision and recall, averaged 97.7%. In contrast, the VOG system demonstrated lower and more variable performance across five test iterations, with an average accuracy of 82.6% (SD=10.4%), precision of 84.0%, recall of 84.4%, and an F1-score of 82.2%.

Beyond detection accuracy, the computational requirements of both systems showed substantial differences. The EOG system operated at an average of 47.97 FPS, while the VOG system achieved only 5.28 FPS. Memory utilization showed even greater disparities, with the EOG system requiring only 0.69 MB of physical memory compared to the VOG system's 228.38 MB. Similar efficiency gaps were observed in virtual memory usage, where EOG consumed 0.91 MB versus VOG's 353.14 MB. These findings are summarized in Table 2 and Table 3, which compare memory usage and processing speed across both high-performance workstation and embedded hardware platforms; the embedded results will be discussed in the following section.

The performance disparity became even more pronounced for blink detection tasks (Fig. 17). The EOG system achieved near-perfect detection with an average accuracy of 97.3% (SD=2.8%), precision of 97.3%, and perfect recall of 100% across all tests, resulting in an average F1-score of 98.6%. The VOG system, however, exhibited significantly lower and highly variable performance for blink detection, with accuracy ranging from 68% to 88% (mean=77.2%, SD=8.8%). While the VOG system often achieved high precision (average of 96.4%), its recall was consistently lower (mean=79.6%), resulting in an average F1-score of 87.0%. These results indicate that under normal conditions, the EOG-based approach provides substantially more reliable and consistent performance for both command selection and blink detection compared to the VOG system.

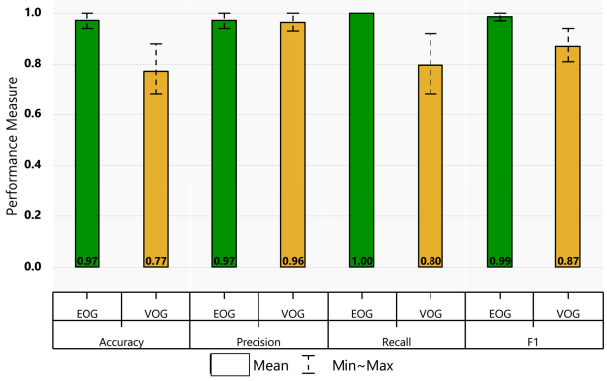


FIGURE 17. Blink comparison under normal condition.

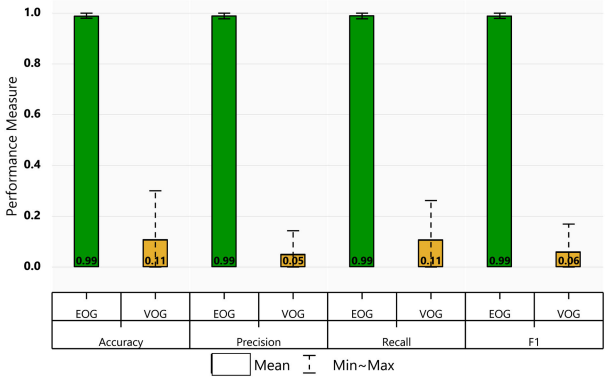


FIGURE 18. Command selection comparison under low-light condition.

TABLE 2. Comparison of memory usage for EOG and VOG across different hardware platforms.

Platform	Memory Type	EOG (MB)	VOG (MB)
Workstation	Physical	0.69	228.38
	Virtual	0.91	353.14
NVIDIA Jetson Orin Nano	Physical	2.18	725.70
	Virtual	66.34	1536.00

TABLE 3. Comparison of processing speed (FPS) between EOG and VOG on different hardware platforms.

Platform	EOG (FPS)	VOG (FPS)
High-Performance Workstation	47.97	5.28
NVIDIA Jetson Orin Nano	38.87	2.80

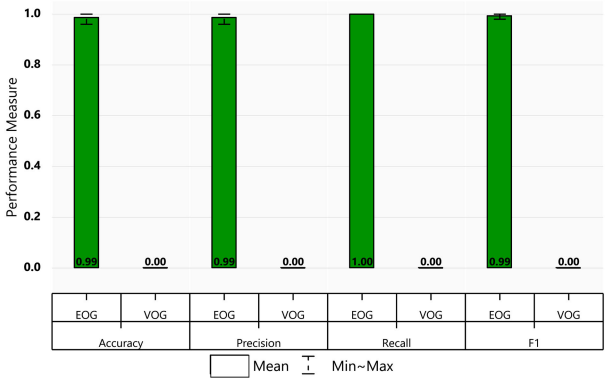


FIGURE 19. Blink comparison under low-light condition.

2) LOW-LIGHT CONDITIONS

To assess the robustness of each approach when visual detection becomes challenging, we evaluated system performance under low-light conditions. Tests were conducted at illumination levels of 0.4, 1.6, and 27 lux to represent a range of lighting scenarios that might be encountered in real-world applications.

The EOG-based system maintained remarkably consistent performance across all low-light conditions. For command selection, the EOG system showed no degradation compared to normal lighting conditions, with average accuracy of 98.8% (SD=1.1%), precision of 98.9%, recall of 98.9%, and F1-score of 98.8% (Fig. 18). These results demonstrate that the electrode-based approach is entirely unaffected by ambient lighting changes. In stark contrast, the VOG system exhibited severe performance deterioration as illumination decreased. At 27 lux, command selection accuracy dropped dramatically to 30% (SD=15.9%), while at the lower illumination levels of 1.6 lux and 0.4 lux, the system performed near chance levels with accuracies of just 2% and 0% respectively. F1-scores followed a similar pattern, declining from 16.9% at 27 lux to effectively 0% at the lowest illumination levels.

For blink detection, the performance disparity was even more pronounced (Fig. 19). The EOG system maintained perfect or near-perfect performance (average accuracy 98.7%) across all low-light conditions, with F1-scores averaging 99.3%. The VOG system, however, completely failed to detect blinks at all low-light conditions, with 0% accuracy across all metrics. This indicates a complete breakdown of the visual detection capabilities in low-light environments.

These results demonstrate the fundamental advantage of LSTM-based EOG eye tracking in low-light environments, where camera-based approaches face insurmountable limitations due to insufficient illumination for proper image capture and feature extraction. Since EOG approach relies on biosignals, it continues to function at full capacity regardless of ambient lighting conditions, making it significantly more versatile for real-world applications where lighting cannot be controlled.

3) WINDY CONDITIONS

To simulate real-world disturbances that might affect eye tracking performance, we evaluated both systems under windy conditions using a Honeywell Turboforce Power+ fan at maximum speed. In this scenario, the EOG-based system demonstrated complete resilience to the wind disturbance. For command selection, the system maintained

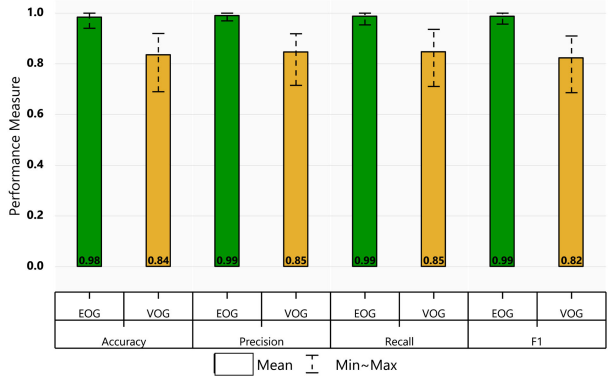


FIGURE 20. Command selection comparison under windy condition.

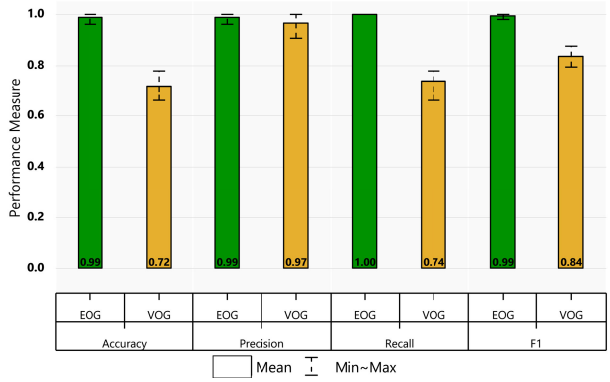


FIGURE 21. Blink comparison under windy condition.

exceptional performance with an average accuracy of 98.4% (SD=2.6%), virtually identical to its performance under normal conditions. Similarly, for blink detection, the EOG system maintained near-perfect performance with an average accuracy of 98.8% (SD=1.7%).

The VOG system maintained consistent performance under windy conditions for command selection, achieving an average accuracy of 83.6% (SD=8.8%), with precision, recall, and F1-scores of 84.7%, 86.1%, and 82.3% respectively. These values are comparable to those observed under normal conditions, indicating that the VOG system was largely unaffected by the wind disturbance (Fig. 20). For blink detection, the VOG system achieved an average accuracy of 72.0% (SD=4.9%) with high precision at 96.6% and an F1-score of 83.7% (Fig. 21).

Overall, both systems demonstrated good resilience to wind disturbances, maintaining performance levels consistent with those observed under normal conditions. This suggests that both EOG and VOG approaches can be effectively deployed in environments with moderate air movement without significant performance degradation.

4) EDGE COMPUTING PERFORMANCE

To assess the feasibility of deploying both eye tracking approaches on resource-constrained hardware, we conducted

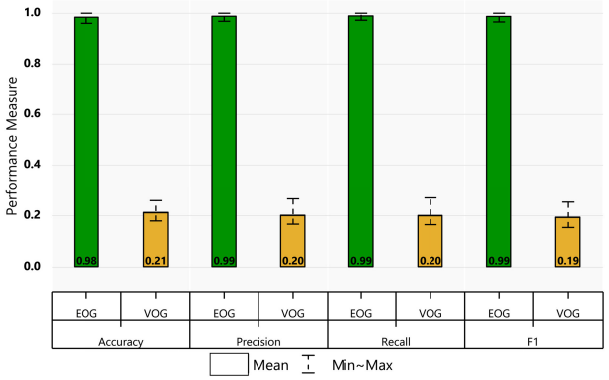


FIGURE 22. Command selection comparison using the NVIDIA Jetson Orin Nano.

performance testing on an NVIDIA Jetson Orin Nano developer kit [19]. This evaluation is critical for applications requiring mobility or deployment in environments where high-performance computing resources are unavailable.

The EOG-based system maintained excellent performance on the edge computing platform (Fig. 22). For command selection, the system achieved an average accuracy of 98.4% (SD=1.5%), with precision, recall, and F1-scores of 98.8%, 98.9%, and 98.7% respectively. These results are virtually identical to those obtained on the high-performance workstation, indicating that the EOG approach is highly suitable for edge deployment. The EOG system maintained a consistent frame rate of approximately 38.9 FPS with minimal memory impact, utilizing only 2.2 MB of physical memory on average. (Refer to Table 2 and Table 3 for a summary of platform-specific performance metrics.)

In stark contrast, the VOG system experienced severe performance degradation when deployed on the NVIDIA Jetson Orin Nano platform. Command selection accuracy dropped dramatically to an average of 21.2% (SD=3.7%), with corresponding precision, recall, and F1-scores of 20.1%, 20.0%, and 19.3% respectively. This significant reduction in performance can be attributed to the computational demands of the computer vision algorithms, which could not maintain real-time processing on the resource-constrained hardware. The VOG system achieved only 2.8 FPS on average, well below the threshold required for responsive interaction, while consuming approximately 725.7 MB of physical memory.

For blink detection, the performance disparity was similarly pronounced. The EOG system maintained near-perfect performance with an average accuracy of 98.1% (SD=2.4%) and an F1-score of 99.0%. The VOG blink detection system, however, achieved an accuracy of only 48.0% (SD=9.7%). While the VOG system maintained perfect precision, its poor recall (48.0%) resulted in an F1-score of 64.4% (Fig. 23).

These results highlight a critical advantage of the EOG approach for edge computing applications: its significantly lower computational requirements enable high-performance operation on resource-constrained hardware, making it well-suited for mobile or embedded applications where power and computational resources are limited.

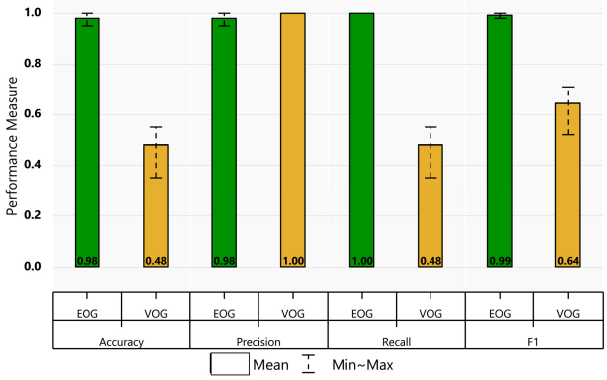


FIGURE 23. Blink comparison using the NVIDIA Jetson Orin Nano.

C. REAL-TIME DRONE CONTROL ON EMBEDDED PLATFORM

In the real-time setup, the system featured a GUI displaying 15 command buttons on a monitor with physical size of 32.15 × 14.38 inches. These commands enabled comprehensive drone control, including basic movements (lift, land, right, left, forward, backward, upward, downward), flip maneuvers (right, left, forward, backward), and rotational actions (clockwise and counterclockwise). For instance, the lift command initiated takeoff, while land brought the drone down safely. Movements in various directions, such as right, left, forward, and backward, enabled lateral and linear control. Upward and downward commands adjusted the altitude, while flipping commands allowed the drone to perform aerial flips in different directions. Rotational commands, clockwise and counterclockwise, handled drone orientation.

When the user focused their gaze on a particular command in the GUI for three seconds, a voice prompt would confirm if the selected command was intended. Upon detecting a blink from the user, the system sent the command to the DJI Tello drone [20] for execution. This interaction ensured that the drone responded only to intentional commands, minimizing the risk of accidental actions.

Figs. 24, 25, and 26 showcase the system in action by capturing the GUI interface, the user's eye movements as they select commands, and the drone's corresponding responses before and after executing the lift, left movement, and flip right commands, respectively. These results illustrate the practical application of our system in a real-time environment, and its performance can be further explored through the provided video link.

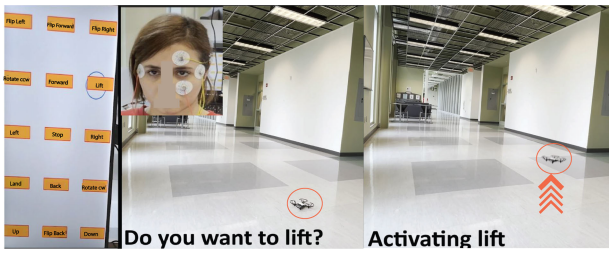


FIGURE 24. Lift command execution.

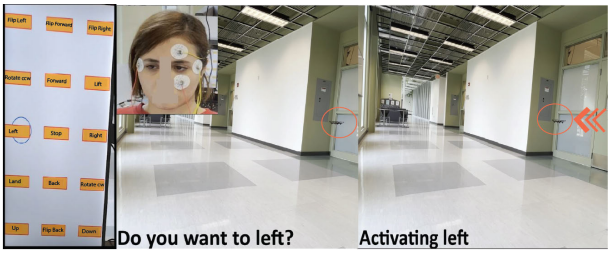


FIGURE 25. Left command execution.



FIGURE 26. Flip right command execution.

IV. DISCUSSION

Through systematic evaluation across diverse conditions, we have demonstrated that our LSTM-based EOG system offers substantial advantages in accuracy, environmental resilience, and computational efficiency compared to VOG. Under normal conditions, the EOG system achieved 97.6% command selection accuracy and 97.3% blink detection accuracy, compared to VOG's 82.6% and 77.2% respectively. The computational requirements showed even greater disparities, with EOG operating at 47.97 FPS while consuming only 0.69 MB of memory, compared to VOG's 5.28 FPS and 228.38 MB.

The significance of this comparison becomes evident when considering previous studies on eye-based UAV control systems, which have established VOG-based control as a viable alternative to manual methods for UAV navigation, with findings demonstrating that VOG reduces trajectory length by 18% compared to manual controls. Assessments revealed that although users reported higher physical and mental demand with VOG interfaces, they experience less frustration and perceive better performance compared to manual controls [38]. This creates a clear progression: if VOG improves upon manual control, and our research shows that EOG outperforms VOG, then EOG represents an advancement over both traditional manual and VOG-based control methodologies for UAV navigation.

Our LSTM-based EOG approach also demonstrates significant improvements over existing EOG-based drone control systems. Previous EOG implementations relied on threshold-based classification methods, which fundamentally lack the ability to capture temporal patterns in eye

movement signals. These systems achieved 94.8% accuracy for gaze direction classification but were limited to only four commands (left, right, up, down), with system delays around 0.6 s [13]. Our LSTM architecture directly addresses these limitations by modeling the temporal dependencies in EOG signals, enabling both higher accuracy (97.6%) across nine distinct commands and faster response times (21 ms) using parallel threading architecture. This temporal modeling is particularly effective for blink detection, which consists of characteristic positive-negative spike patterns that sequential models can better recognize.

While our EOG-based approach has proven effective for UAV control, its application across different robotic platforms presents opportunities and challenges for future work. The current system translates eye movements and blinks into drone navigation commands, but this mapping would require reconfiguration for other types of robots with different movement capabilities. Ground vehicles, humanoid robots, and industrial robots would each need a customized command interface, but the underlying EOG signal processing and LSTM architecture could remain unchanged. This inherent flexibility suggests broad applicability of the described approach, though platform-specific implementation and testing would be necessary to validate performance across diverse robotic systems.

Our testing has validated the system's resilience to environmental variations, maintaining accuracy above 98% under low-light conditions and wind disturbances. This stability in challenging environments demonstrates the inherent advantage of bioelectric signal processing over optical methods for gaze tracking. This environmental robustness suggests broad applicability, though additional testing in scenarios involving natural and fatigue user movements would provide further validation of the system's performance during typical HRI scenarios.

The computational efficiency of our approach is evident from our analysis, indicating minimal memory usage and strong processing performance. These metrics demonstrate the lightweight nature of our EOG processing pipeline, particularly when compared to vision-based alternatives that typically require much more computational resources. The consistent performance across high-performance and embedded platforms suggests that our system could operate efficiently on even more constrained devices. While these computational metrics strongly indicate energy-efficient operations suitable for battery-powered applications, systematic power consumption testing would provide more direct evidence of the system's efficiency and potential operational duration on various platforms.

Building on these insights, our future work includes expanding implementation across different robotic systems, evaluating performance during natural and fatigue user movements, and exploring user personalization through transfer learning techniques. Additionally, power consumption analysis could provide further optimization opportunities for extended operations. These expansions would build upon the foundations established in the work described in the present paper and broaden the system's application domains.

V. CONCLUSION

This study introduced a real-time, hands-free UAV control system based on EOG signals, using a dual-output LSTM model to detect gaze-based commands and blinks. We implemented a vision-based VOG system as a baseline and conducted a comparative study under various environmental and hardware conditions. The EOG-based system achieved 97.6% command selection accuracy and 97.3% blink detection accuracy under normal conditions, significantly outperforming the VOG system, which reached 82.6% and 77.2%, respectively. These advantages persisted in challenging environments, with the EOG system maintaining over 98% accuracy under low-light conditions, where the VOG system experienced a notable drop in performance.

In addition to accuracy, the system offered substantial computational advantages. It ran at 47.97 FPS using only 0.69 MB of memory on a high-performance workstation and maintained 38.87 FPS on the resource-constrained NVIDIA Jetson Orin Nano. Compared to the VOG system, this represents a 13.9× improvement in speed and a 319× reduction in memory usage, highlighting its suitability for embedded and real-time applications.

These findings demonstrate that biosignal-based gaze tracking not only is technically feasible but also provides measurable advantages in accuracy, responsiveness, and computational efficiency. This positions the EOG approach as a practical alternative to vision-based VOG systems, particularly in environments demanding real-time performance and robust adaptability.

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